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Influence of browsing damage and overstory cover on regeneration of American beech and sugar maple nine years following understory herbicide release in central Maine

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Abstract Northern hardwood stands, notably those with American beech (*Fagus grandifolia* Ehrh), sugar maple (*Acer saccharum* Marsh.), and yellow birch (*Betula alleghaniensis* Britton), are abundant across the forested landscapes of northeastern USA and southeastern Canada. Recent studies have reported an increasing dominance of American beech in the understory and midstory of these forests. Beech is a commercially less desirable tree species due to its association with beech-bark disease, and because it commonly interferes with the regeneration of other more desirable tree species. We examined hardwood regeneration characteristics nine years after application of a 3 $\times$ 4 factorial combination of glyphosate herbicide (0.56, 1.12, and 1.68 kg ha^{-1}) and surfactant concentrations (0.0, 0.25, 0.5, and 1.0% v v^{-1}) to release sugar maple regeneration from beech-dominated understories using three stands that received shelterwood seed cutting in central Maine. Measurements nine years after treatment showed that glyphosate rate increased both absolute (*AD*) and relative density (*RD*) of sugar maple regeneration, but not its height (*HT*). In contrast, beech *AD*, *RD,* and *HT* were all significantly reduced with increasing glyphosate rate. Post-release browsing by ungulates and a high residual

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overstory basal area resulted in reduced sugar maple *HT*. Our results indicated that glyphosate herbicide applied in stands that have been recently shelterwood seed cut can significantly increase the abundance of sugar maple regeneration. However, subsequent browsing damage combined with the negative influence of the residual overstory cover can limit the longer-term benefit of understory herbicide treatments. Subsequent removal of the overstory and browsing-control measures may be needed to promote sugar maple regeneration over beech in similar northern hardwood stands.

Keywords Northern hardwoods of USA · Natural regeneration · Glyphosate · Surfactant · Browsing · Canopy influences · Vegetation management · Beech control

Introduction

Northern hardwood forests, notably those with American beech (*Fagus grandifolia* Ehrh), sugar maple (*Acer saccharum* Marsh.), and yellow birch (*Betula alleghaniensis* Britton), are abundant across the forested landscape of northeastern USA and southeastern Canada (Angers et al. 2005; Bose et al. 2017a), and provide an important source of sawlogs and fiber for the wood products industry. Recent studies have reported increasing dominance of American beech in the understory and midstory of these mixed hardwood forests (e.g., Hane 2003; Beaudet and Messier 2008; Duchesne and Ouimet 2009; Bose et al. 2017b), as well as interference by beech that limits the regeneration of other more desirable tree species (Nyland et al. 2006a, b). Beech was historically considered to be a commercially desirable hardwood species before beech-bark disease substantially lowered the wood quality and growth of infected trees (Houston 1975; Morin et al. 2007; Kasson and Livingston 2012). With progression of the disease, understory root suckers increased in many stands, which interfered with the regeneration of other species. Today, finding effective ways to reduce these dense sub-canopy layers of root suckers, and promote the establishment of other species presents a major challenge to maintaining the diverse and productive composition within this forest type (Hane 2003; Nyland et al. 2006a).

In northern hardwood forests, the combination of a long fire return-interval and repeated partial harvesting as the primary forest disturbance factors have favored the dominance of shade tolerant tree species (including beech and sugar maple) in the forest understory (Seymour 1995; Nolet et al. 2008; Bose et al. 2016). Of these species, beech seedlings and root suckers have shown a higher survival and growth rate than sugar maple under low-light conditions (Kobe et al. 1995; Beaudet and Messier 1998). Limited light availability is common in understories of northern hardwoods due to partial harvesting and the increased presence of root suckers that have followed the spread of beech-bark disease (Nyland et al. 2006a; Runkle 2007). Moreover, unlike sugar maple, beech regenerates vegetatively from root suckers (Beaudet and Messier 1998), often after damage to their shallow roots during harvesting operations and other disturbances (Jones and Raynal 1988; Farrar and Ostrofsky 2006).

In addition, browsing ungulates, notably white-tailed deer (*Odocoileus virginianus* Zimmerman) and moose (*Alces alces* L.), prefer maple and birch species over beech, which leads to higher survival and better development of beech regeneration than other tree species (Beaudet and Messier 1998). Many studies have highlighted the role of selective browsing by ungulate herbivores in changing tree species composition in the forested landscape across the world (e.g., Webb et al. 1956; Farnsworth and Richards 1971; Rooney 2001; Sage et al. 2003; Beguin et al. 2009; Rooney et al. 2015). Therefore, a key challenge for managing regeneration in these mixed-hardwood forests is taking deliberate measures to control ungulate damage to high-value hardwood species (i.e., maple and birch), and to reduce understory beech interference.

Pre-harvest site preparation (Kelty and Nyland 1981) and post-harvest herbicide application (Ostrofsky and McCormack 1986; Wagner et al. 2010) have been suggested as methods to reduce beech interference. Initial (1–3 year post-herbicide application) results indicated that glyphosate herbicide can effectively reduce the density of regenerating beech, while preserving or increasing the relative densities of more commercially desirable species such as sugar maple, red maple (*Acer rubrum* L.), and yellow birch (Ostrofsky and McCormack 1986; Nelson and Wagner 2011). However, limited information currently exists on the medium-term (5–10 year) response of hardwoods stands to post-cutting glyphosate application (Wagner et al. 2010). In particular, key questions include: (1) whether the initial success of herbicide application is sustained over the longer term, (2) whether the low palatability of beech to browsers influences subsequent regeneration dynamics, and if this limits the early benefits of herbicide application, and (3) whether shading by the residual overstory canopy reduces the effect of earlier herbicide treatment, as beech is more shade tolerant than other hardwood tree species (Brisson et al. 1994; Nyland et al. 2006a; Wagner et al. 2010).

Nelson and Wagner (2011) documented a positive effect of different glyphosate and surfactant concentrations in shifting the dominance of understory regeneration from beech to sugar maple three-years after treatment at three sites in central Maine. We examined whether these early outcomes persisted through nine years after treatment, and the longer-term effects of glyphosate herbicide rates, surfactant concentration, degree of browsing damage, and overstory basal area on the density and height of understory tree species. We tested three primary hypotheses: (1) increasing glyphosate rate and surfactant concentration reduced the density and height of beech regeneration, and concomitantly increased the density and height of sugar maple regeneration for the first decade after treatment (Ostrofsky and McCormack 1986; Nelson and Wagner 2011); (2) if present, browsing pressure confounded the longer-term benefits of herbicide treatment to control beech and favor sugar maple (Tripler et al. 2005; Wagner et al. 2010); and (3) the degree of residual crown closure following shelterwood seed cutting would increase the density and height of beech regeneration relative to that of more desirable tree species (Beaudet and Messier 1998; Beaudet et al. 1999).

Methods

Study sites

Three sites (T2-R7, T2-R8, and TA-R7) with hardwood stands that had beech and sugar maple dominated understories were selected in north-central Maine, USA. The three sites are all within ~32 km of Millinocket, Maine. Locations 1, 2, and 3 were located at 45°49′N, 68°33′W, 45°47′N, 68°42′W, and 45°35′N, 68°36′W, respectively (Fig. 1). The climate at Millinocket is continental with mean annual precipitation of 1058 mm evenly distributed throughout the year (Baron et al. 1980). The mean annual temperature is 5.3 °C associated with mean monthly temperature in January and in July is −10.0 and 19.8 °C, respectively (Baron et al. 1980). The soils of the region are of glacial till origin and

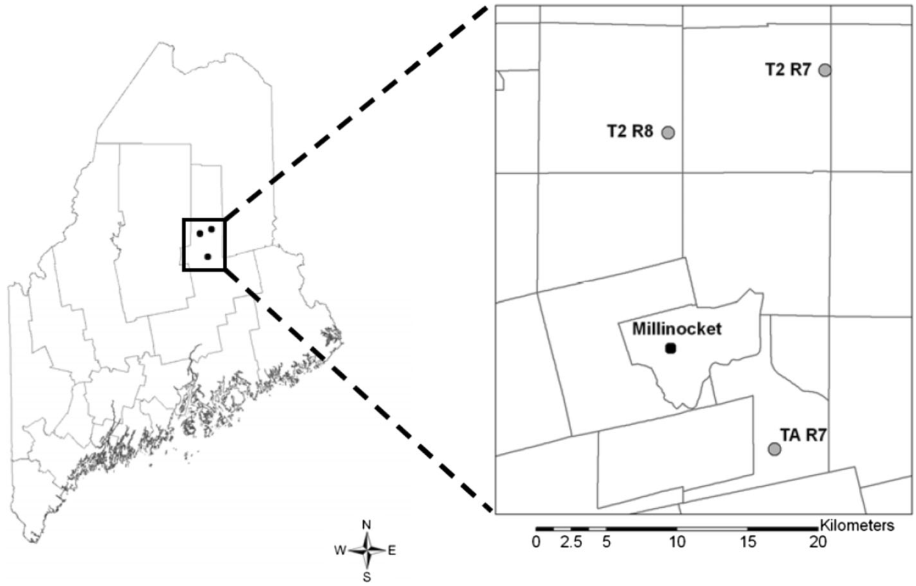

Fig. 1 Locations of three study sites (T2R7, T2R8, and TAR7) in north-central Maine. All three sites are located within 20 km of Millinocket, Maine, USA

characterized by Typic Haplorthods stony sandy–silty loams with slopes between 0 and 15% (USDA Natural Resources Conservation Service 2008).

Harvesting treatments

Shelterwood seed cutting occurred at all three study sites from 2002 to 2004 before this experiment was installed. The objective of the shelterwood seed cut was to stimulate the growth of understory hardwood regeneration by removing adequate basal area from the overstory and midstory. Pre-harvest basal area of three sites ranged from 30 to 34 m^2 ha^{-1} and was between 9 and 20 m^2 ha^{-1} following harvesting (Table 1; Nelson and Wagner 2011). The shelterwood seed cutting resulted in high densities of beech, sugar maple, red maple, striped maple (*Acer pensylvanicum* L.), and yellow birch regeneration (Nelson and Wagner 2011).

Herbicide treatments

In August 2006, a 3 × 4 factorial combination of glyphosate herbicide (Accord® Concentrate) and surfactant concentrations (Entrée 5735®) were tested. These factors included glyphosate rates of 0.56, 1.12, and 1.68 kg ha^{-1} and surfactant concentrations of 0.0, 0.25, 0.5, and 1.0% (v v^{-1}) plus a non-treated control. Each treatment was applied in aqueous solution at 93.5 L ha^{-1} using a backpack sprayer. Treatments were applied using two identical CO_2 pressurized backpack sprayers equipped with a KLC-9 Floodjet nozzle attached to the end of a 3.35-m-long boom (R&D Sprayers, Inc., model 4F). All herbicide and surfactant mixtures were prepared in 2-L bottles in a laboratory before application. Both sprayers were calibrated prior to application, and walking time across plots were monitored using a stopwatch to ensure precise application. The swath width of each

Table 1 Pre- and post-harvest stand structural and compositional characteristics of three study sites

Characteristics	Pre-harvest				Post-harvest			
	Site-1: T2-R7	Site-2: T2-R8	Site-3: TA-R7	Mean ± SD	Site-1: T2-R7	Site-2: T2-R8	Site-3: TA-R7	Mean ± SD
Tree density (stems ha^{-1})	1306.9	1151.0	1441.2	1299.7 ± 145.2	937.4	677.0	1041.6	885.4 ± 187.8
Quadratic mean dbh in cm	17.3	18.2	17.1	17.5 ± 0.6	15.9	13.0	15.8	14.9 ± 1.6
Overstory basal area of American beech (m^2 ha^{-1})	6.0	24.6	5.5	12.0 ± 10.9	1.5	5.9	5.5	4.3 ± 2.4
Overstory basal area of sugar maple (m^2 ha^{-1})	5.9	3.0	13.9	7.6 ± 5.6	4.3	3.0	4.2	3.8 ± 0.7
Overstory basal area of other hardwoods (m^2 ha^{-1})	14.1	0.1	5.6	6.6 ± 7.1	10.1	0.1	3.0	4.4 ± 5.1
Overstory basal area of softwoods (m^2 ha^{-1})	4.7	2.3	8.1	5.0 ± 2.9	2.7	0	7.7	3.5 ± 3.9
Total overstory basal area (m^2 ha^{-1})	30.7	30.0	33.1	31.3 ± 1.6	18.6	9.0	20.4	16.0 ± 6.1

Trees >4.0 cm at dbh were only considered sprayer was equal to the width of the treatment plot. Each sprayer was calibrated to deliver half of the target delivery rate, so that each plot could be sprayed in two passes in opposite directions to ensure good coverage of the entire plot. The sprayers were rinsed with water between treatment applications. The volume of unused spray mixture remaining after treating each plot was measured to calculate the volume of herbicide (kg ha^{-1}) and surfactant (L ha^{-1}) applied to a treatment plot. The average temperature during treatment application was 24 °C, the average relative humidity was 44%, and the average wind speed was 2.6 km h^{-1}. No rain occurred for at least a week after application.

Experimental design

In November 2015 (nine years after herbicide application), all 39 of the original treatment units (13 treatment combinations × 3 locations) of Nelson and Wagner (2011) were relocated. Due to the increased size of understory trees since earlier measurements, a larger measurement plot than that of Nelson and Wagner (2011) was needed. Therefore, two 8 × 2 m (16 m^2) measurement plots were nested within each 36 m × 7.6 m (273. 6 m^2) treatment plot, but were positioned to include all original sample plots used by Nelson and Wagner (2011) to ensure consistency of measurement. All plots were placed to avoid logging trails. One treatment plot at Site 1 had been accidentally destroyed since the last measurement due to road construction and was not available for this analysis.

Field methods

Stem diameter (measured in 2 cm classes at 50 cm aboveground), total height (nearest dm), and species of all live understory trees (>50 cm in height and <10 cm dbh) were measured and recorded for each measurement plot. Browsing damage (all by deer and/or moose) on all measured trees was assessed and recorded: (1) non-browsed (no sign of browsing), (2) lightly browsed (2–3 defects; i.e., crooks, forked, and/or pruned), (3) moderately browsed (4–5 defects), and (4) heavily browsed (pruned and >5 defects), and the number of stems per browse class in each plot was converted to stems ha^{-1}. Overstory trees ($\geq$10 cm dbh) were assessed using a variable-radius plot with BAF (basal area factor) prism of 2.296 m^2 ha^{-1} at the center of each measurement plot by recording dbh (nearest cm) and species.

Analytical approach

Response variables

Based on the hypotheses and number of stems by species, regeneration (0.5–4.9 m in height) was categorized into four species groups: (1) beech, (2) sugar maple, (3) other hardwoods (all hardwood regeneration except beech and sugar maple), and (4) softwoods. The Softwood group included balsam fir (*Abies balsamea* L.), eastern hemlock (*Tsuga canadensis* (L.) Carr.), northern white cedar (*Thuja occidentalis* L.), red spruce (*Picea rubens* Sarg.), and white pine (*Pinus strobus* L.). The Other Hardwoods group included eastern hophornbeam (*Ostrya virginiana* (Mill.) K. Koch), black cherry (*Prunus serotina* Ehrh.), American hornbeam (*Carpinus caroliniana* Walt.), mountain birch (*Betula cordifolia* (Regel) Fern.), pin cherry (*P. pensylvanica* L. f.), quaking aspen (*Populus tremuloides* Michx.), red maple, striped maple, white ash (*Fraxinus americana* L.), and yellow birch.

Three response variables were created from measurement of the understory regeneration: (1) absolute density (*AD;* total number of stems for each species group in each measurement plot), (2) relative density (*RD*; percentage of stems in each species group relative to the total number of stems in a plot), and average height (*HT*) of all seedlings and saplings by species group in each measurement plot.

Explanatory variables

To test the three hypotheses, three explanatory variables were used. Measured volumes of glyphosate (kg ha^{-1}) and surfactant (L ha^{-1}) delivered to each treatment plot were used as primary explanatory variables given the original experimental design. Levels of browse damage were used as the second explanatory variable. Given that only 20 of 4529 measured stems showed no sign of animal browsing, the unbrowsed trees were excluded from the final analysis due to small sample size. Overstory tree basal area was considered as an indicator of canopy crown closure. Since overstory tree basal area was highly correlated with overstory tree density, only basal area was used in the analysis. In addition, preliminary analysis showed that quadratic mean dbh, and the ratio between overstory hardwood to softwood basal area, were weaker predictors than overstory tree basal area, and were therefore also not included in the final analysis.

Because the levels of overstory basal area and browsing damage were not systematically controlled in the subsequent analysis of this experiment, we examined whether the range of observations of overstory and browsing were fully represented in all combinations among the experimental units, as well as whether there was any relationship with either of the original controlled factors (herbicide rate and surfactant concentration). We assessed scatter distributions of the predictor variables (see Supplemental Materials section) and found no apparent systematic relationship among any of the four predictor variables, and that there was sufficient representation of all combinations of the predictor variables among the experimental units. We addressed the large variability of regeneration density across measurement plots by incorporating the associated random effects of the nested design (plots nested within experimental units, and experimental units nested within sites) into the model form, and by including a null model (i.e., no effect of explanatory variables on the response variables) in the list of candidate models.

Statistical analysis

Ten models were considered based on the biological hypotheses for the three response variables, absolute density (*AD*), relative density (*RD)*, and height of regeneration (*HT*) (Table 2). A null model was included to test the hypothesis of no effect of explanatory variables on the response variable. Eight models incorporated the linear effects of herbicide rate, surfactant concentration, browsing (three levels), overstory basal area, species types (four levels), and various two-way interactions for each response variable (Table 2). A global model was generated by incorporating the effects of all explanatory variables (Table 2). Candidate models were compared using Akaike's Information Criterion corrected for small samples (AICc). Multi-model inferences were used to compute the model-averaged estimates of the explanatory variables and their 95% confidence intervals (Burnham and Anderson 2002) using the AICcmodavg package in R (Mazerolle 2011). A confidence interval excluding 0 indicates that the response variable varied with the explanatory variables of interest (Burnham and Anderson 2002; Mazerolle 2006).

Table 2 List of candidate models considered in the analysis and biological hypothesis for each response variables (*RV*) tested: regeneration absolute density (*AD*), regeneration relative density (*RD*), and regeneration average height (*HT*)

Models	Candidate models	Model types (biological hypothesis)
1	RV ~ 1	Null model
2	RV ~ RATE * SPP	Positive effect of herbicide rate but the effect changes according to species types
3	RV ~ SURF * SPP	Positive effect of surfactant but the effect changes according to species types
4	RV ~ RATE * SPP + SURF * SPP	Positive effect of herbicide rate with an additive effect of surfactant but the effect changes according to species types
5	RV ~ BROWSE * SPP	Negative effect of browsing but the effect changes according to species types
6	RV ~ BA * SPP	Positive effect of overstory basal area but the effect changes according to species types
7	RV ~ RATE * SPP + SPP * BROWSE + RATE * BROWSE	Seedling density of different species varies by herbicide rate and browsing classes, and also by the interaction between herbicide rate and browsing classes
8	RV ~ RATE * SPP + SPP * BA + BA * RATE	Seedling density of different species varies by herbicide rate and overstory basal area, and also by the interaction between herbicide rate and overstory basal area
9	RV ~ BROWSE * SPP + SPP * BA + BA * BROWSE	Seedling density of different species varies by overstory basal area and browsing classes, and also by the interaction between browsing classes and overstory basal area
10	RV ~ RATE * SPP + SURF * SPP + BROWSE * SPP + BA * SPP + BA * BROWSE + BA * RATE + RATE * BROWSE	Global model

Herbicide rate (RATE), surfactant concentration (SURF), species groups (SPP), degree of browsing (BROWSE), and overstory basal area (BA). Four species were considered in the analysis including American beech, sugar maple, other hardwoods, and softwood species

The interaction term between explanatory variables was indicated by a * sign

Effects of explanatory variables on *AD* and *RD* were assessed by generalized linear mixed-effect models using the function glmer in the lme4 package in R (version 3.2.5), where Poisson and binomial families were used, respectively. *AD* was analyzed with original count data. The effect of explanatory variables on *HT* were assessed by linear mixed-effect models using the function lme of the nlme package in R (Pinheiro et al. 2014; R Development Core Team 2014). Preliminary analysis indicated that an additional error structure to account for plot spatial autocorrelation did not improve model performance and was not incorporated in the final model. To improve model fit, a square root transformation was used for *HT*. We visually verified the assumptions of normality and variance homogeneity of the residuals. We also calculated adjusted R^2 and % RMSE (root mean square error) for each model. The adjusted R^2 of the mixed-effect model was calculated using the function of "r.squaredLR" of muMIn package in R (Bartoń 2013; R Development Core Team 2014). The partial contribution to R^2 by each predictor variable was quantified by using their respective sum of squares in the ANOVA. For all models, measurement plots nested within treatment units, and treatment unit nested within sites were treated as random effects.

Results

Nine years after treatment, absolute density (*AD*), and relative density (*RD*) of understory sugar maple was higher than all other species groups, and was $\sim$2.5 times higher than beech (Table 3). However, data revealed a high degree of variability in regeneration density among species across plots (Table 3), and was a factor that we needed to accommodate in the statistical analyses. The regeneration height (*HT*) of sugar maple was lower than *HT* of all other species (Table 3). The relatively lower *HT* of sugar maple was partially due to browsing by deer and/or moose, as approximately 67% of sugar maple seedlings across the study were heavily browsed; which was higher than all other species (Tables 3, 4). The *AD* and *RD* of other hardwoods (dominated by red maple and yellow birch) were also higher than beech, but were not as severely browsed as sugar maple (Tables 3, 4). The *AD* and *RD* of softwoods were the lowest among the four species groups (Table 3).

Regeneration absolute density (*AD*)

Among the 10 models predicting *AD*, the most complex model incorporating the additive effects of all explanatory variables and the two-way interactions, had the highest support of AICc, as well as the highest adjusted R^2 (Table 5). The effect of glyphosate rate, surfactant concentration, and overstory basal area on *AD* varied by species group (i.e., interaction had strong effect) (Table 6). For example, the *AD* of sugar maple and softwoods were not affected by the surfactant concentration, but the *AD* of beech and other hardwoods decreased with increased surfactant concentration (Table 6; Fig. 2a–c). Increased glyphosate rate increased the *AD* of sugar maple and softwoods, and reduced the *AD* of beech and other hardwoods (Table 6; Fig. 2a). However, increasing overstory basal area only increased the *AD* of softwoods, but not the *AD* of other species (Table 6; Fig. 2c).

Regeneration relative density (*RD*)

The model that incorporated the additive effects of glyphosate rate, browsing class, and species, as well as the two-way interactions among those variables, had the highest support of AICc and the highest R^2 (Table 5). As with the results for *AD*, the effect of glyphosate rate, surfactant concentration, and overstory basal area on *RD* was dependent on species group (interaction had strong effect) (Table 6). For example, the *RD* of softwoods and beech were not influenced by surfactant concentration, but the *RD* of other hardwoods was reduced by increasing surfactant concentration, while sugar maple *RD* increased with surfactant concentration (Table 6; Fig. 2e). As with results for *AD*, increasing glyphosate rate increased the *RD* for sugar maple and softwoods, but reduced the *RD* of beech and

Table 3 Characteristics of response variables nine years after glyphosate treatments across three locations in north-central Maine (total number of plots 77)

Characteristics	Mean	SD	Range
Absolute density (stems ha^{-1})			
American beech	4910.7	6550.6	0.0–31,875.0
Sugar maple	13,076.3	15,544.8	0.0–80,625.0
Other hardwoods	7102.3	6039.1	0.0–32,500
Softwoods	3668.8	5215.7	0.0–26,875
Relative density (% of total)			
American beech	16.2	19.2	0.0–82.6
Sugar maple	39.9	28.0	0.0–94.2
Other hardwoods	25.1	17.2	0.0–80.6
Softwoods	18.8	25.0	0.0–95.5
Average height (m)			
American beech	1.81	1.01	0.50–4.80
Sugar maple	1.36	1.15	0.50–4.90
Other hardwoods	1.84	1.43	0.50–6.40
Softwoods	1.52	0.91	0.50–4.30

other hardwoods (Table 6; Fig. 2d), whereas higher overstory basal area slightly increased the *RD* of softwoods. The *RD* of sugar maple, beech, and other hardwoods was not influenced by overstory basal area (Table 6; Fig. 2f).

Regeneration height (*HT*)

Among the 10 models of *HT*, the most complex model that incorporated the additive effects of all explanatory variables, as well as various two-way interactions, had the full support of AICc and the highest R^2 (Table 5). Overstory basal area had a stronger association with *HT* than glyphosate rate, surfactant concentration, or browsing damage classes (Table 7). The effect of glyphosate rate and overstory basal area on *HT* was dependent upon species group (i.e., interaction had strong effect), but not upon browsing damage classes (i.e., interaction had no significant effect) (Table 6). The *HT* of beech, sugar maple, and other hardwoods decreased with increased glyphosate rate (Fig. 2g) and surfactant concentration (Fig. 2h), but *HT* of softwoods was not sensitive to glyphosate rate or surfactant concentration. The *HT* of all sugar maple and other hardwoods decreased with higher overstory basal area, but *HT* of American beech and softwood increased with higher overstory basal area (Fig. 2i).

Discussion

The primary objective of this study was to evaluate whether the effects of glyphosate rate and surfactant concentration, as documented three-years post-treatment (Nelson and Wagner 2011), were sustained for the first decade. A secondary objective was to assess whether subsequent browsing damage and overstory closure had influenced the longer-term effects of glyphosate herbicide. Our results indicated that understory release using glyphosate herbicide effectively increased the abundance of sugar maple regeneration, while reducing the abundance of beech regeneration at nine years after treatment.

Table 4 Characteristics of explanatory variables nine years after glyphosate treatments across three locations in north-central Maine (total number of plots 77)

Characteristics	Mean	SD	Range
Actual glyphosate rate (kg ha^{-1})	1.00	0.53	0–1.79
Actual surfactant concentrations (L ha^{-1})	0.36	0.34	0–0.98
Overstory tree basal area (m^2 ha^{-1})	14.8	9.3	2.3–32.1
Overstory beech basal area (m^2 ha^{-1})	4.9	5.2	0.0–25.3
Overstory sugar maple basal area (m^2 ha^{-1})	1.6	2.5	0.0–11.5
Overstory tree density (stems ha^{-1})	440.2	509.0	0.0–3096.0
Non-browsed regeneration (stems ha^{-1})			
American beech	56.8	271.7	0.0–1875.0
Sugar maple	48.7	300.2	0.0–2500.0
Other hardwoods	40.6	185.2	0.0–1250.0
Softwood	0.0	0.0	0.0
Lightly browsed regeneration (stems ha^{-1})			
American beech	771.1	1105.5	0.0–5000.0
Sugar maple	982.1	1607.7	0.0–9375.0
Other hardwoods	1014.6	1455.3	0.0–6250.0
Softwood	974.0	1879.1	0.0–11,875.0
Moderately browsed regeneration (stems ha^{-1})			
American beech	2849.0	4264.9	0.0–21,250.0
Sugar maple	3336.0	4279.5	0.0–25,625.0
Other hardwoods	3133.1	3401.4	0.0–17,500.0
Softwood	1030.8	1513.3	0.0–7500.0
Heavily browsed regeneration (stems ha^{-1})			
American beech	1233.8	2267.1	0.0–9375.0
Sugar maple	8709.4	11,804.4	0.0–60,625.0
Other hardwoods	2914.0	3030.8	0.0–16,250.0
Softwood	1664.0	2937.8	0.0–15,000.0

However, subsequent browsing damage and residual overstory canopy reduced the longer-term benefit of the release treatments by suppressing the height growth of released tree species.

Role of herbicide

Glyphosate applied shortly after the first shelterwood seed cutting in these hardwood stands was successful in shifting dominance of understory regeneration toward sugar maple and away from beech for nine years following treatment. This result is consistent with our initial hypothesis. Beech density decreased with increasing application rate of glyphosate ($\geq$1.12 kg ha^{-1}) for the first nine years after treatment (Fig. 2 of present study; Fig. 3 of Nelson and Wagner 2011), which indicates a sustained effect of glyphosate in controlling beech regeneration and promoting sugar maple regeneration. A similar result of successful beech control (although short-term) has also been reported by other studies in this region (Pitt et al. 1993) and neighboring regions (Horsley and Bjorkbom 1983; Kochenderfer et al.

Table 5 Results of model selection based on AICc criteria of the most probable models for regeneration absolute density (*AD*), regeneration relative density (*RD*), and regeneration average height (*HT*)

Models	Model	AICc	ΔAICc	AICc weight (w_i)	Adjusted R^2	% RMSE
Regeneration absolute density (AD) (Poisson family)						
10	*AD* ~ RATE * SPP + SURF * SPP + BROWSE * SPP + BA * SPP + BA * BROWSE + BA * RATE + RATE * BROWSE	6036.2	0.0	1.00	0.97	19.1
7	*AD* ~ RATE * SPP + SPP * BROWSE + RATE * BROWSE	6229.0	192.8	0.00	0.96	19.7
9	*AD* ~ BA * SPP + SPP * BROWSE + BA * BROWSE	6313.0	276.8	0.00	0.96	19.5
Regeneration relative density (RD) (binomial family)						
7	*RD* ~ RATE * SPP + SPP * BROWSE + RATE * BROWSE	796.8	0.0	0.95	0.21	13.2
10	*AD* ~ RATE * SPP + SURF * SPP + BROWSE * SPP + BA * SPP + BA * BROWSE + BA * RATE + RATE * BROWSE	804.4	7.6	0.02	0.23	13.6
2	*RD* ~ RATE * SPP	804.8	8.1	0.02	0.16	13.1
Regeneration average height (HT) (square root transformed)						
10	*HT* ~ RATE * SPP + SURF * SPP + BROWSE * SPP + BA * SPP + BA * BROWSE + BA * RATE + RATE * BROWSE	1513.8	0.0	1.00	0.63	4.2
9	*HT* ~ BA * SPP + BROWSE * SPP + BA * BROWSE	1565.0	51.2	0.00	0.57	4.1
7	*HT* ~ RATE * SPP + SPP * BROWSE + RATE * BROWSE	1586.1	72.3	0.00	0.55	4.1

Of the 10 tested models for each response variable (see Table 2), only the top three with the lowest AICc values are presented. AICc: Akaike's Information criterion corrected for small samples, ΔAICc: AICc relative to the most parsimonious model, wi: AICc model weight, which indicates the relative likelihood of a model, Adjusted R^2 = coefficient of determination adjusted for number of predictor variables, and % RMSE = % root mean square error

Herbicide rate (RATE), surfactant concentration (SURF), species groups (SPP), degree of browsing (BROWSE), and overstory basal area (BA). Four species were considered in the analysis including American beech, sugar maple, other hardwoods and softwood species

The interaction term between explanatory variables was indicated by a * sign

Table 6 Estimates and their 95% confidence intervals of the effect of explanatory variables on response variables: regeneration absolute density (*AD*), regeneration relative density (*RD*) and regeneration height (*HT*) based on model averaging

Parameter	Estimates (low 95% CI, up 95% CI)		
	AD	RD	HT
Interaction between glyphosate rate and species groups			
Beech to rate versus sugar maple to rate	**1.39 (1.20, 1.58)**	**1.58 (0.62, 2.54)**	0.41 (−0.01, 0.84)
Beech to rate versus other hardwoods to rate	**0.84 (0.64, 1.05)**	0.60 (−0.49, 1.68)	−0.34 (−0.75, 0.07)
Beech to rate versus softwood to rate	**1.59 (1.35, 1.84)**	**2.47 (1.31, 3.63)**	**0.92 (0.45, 1.39)**
Interaction between surfactants and species groups			
Beech to surfactant versus sugar maple to surfactant	**0.91 (0.62, 1.20)**	**2.13 (0.51, 3.76)**	−0.09 (−0.78, 0.60)
Beech to surfactant versus other hardwoods to surfactant	0.14 (−0.19, 0.47)	0.26 (−1.67, 2.20)	**−0.68 (−1.35, −0.002)**
Beech to surfactant versus softwoods to surfactant	**0.75 (0.39, 1.12)**	**1.91 (0.12, 3.69)**	0.53 (−0.23, 1.29)
Interaction between basal area and species groups			
Beech to basal area versus sugar maple to basal area	**0.03 (0.02, 0.04)**	0.03 (−0.03, 0.08)	**−0.05 (−0.08, −0.03)**
Beech to basal area versus other hardwood to basal area	**0.03 (0.02, 0.04)**	0.02 (−0.04, 0.08)	−0.02 (−0.05, 0.001)
Beech to basal area versus softwood to basal area	**0.06 (0.05, 0.08)**	**0.06 (0.01, 0.13)**	−0.02 (−0.05, 0.006)
Interaction between browsing and species groups			
Heavily browsed beech versus lightly browsed sugar maple	**−1.71 (−2.03, −1.40)**	**−1.70 (−3.08, −0.33)**	**−0.61 (−1.15, −0.07)**
Heavily browsed beech versus lightly browsed other hardwoods	**−0.59 (−0.91, −0.26)**	−0.23 (−1.77, 1.31)	−0.21 (−0.75, 0.33)
Heavily browsed beech versus lightly browsed softwoods	−0.07 (−0.41, 0.27)	−0.33 (−1.81, 1.14)	**−0.70 (−1.30, −0.11)**
Heavily browsed beech versus moderately browsed sugar maple	**−1.80 (−2.02, −1.58)**	**−2.11 (−3.45, −0.77)**	**−0.53 (−1.03, −0.03)**
Heavily browsed beech versus moderately browsed other hardwoods	**−0.76 (−1.00, −0.53)**	−0.75 (−2.27, 0.78)	**−0.50 (−0.99, −0.004)**
Heavily browsed beech versus moderately browsed softwoods	**−1.32 (−1.61, −1.02)**	**−1.86 (−3.44, −0.27)**	**−0.72 (−1.28, −0.16)**
Interaction between browsing and glyphosate rate			
Heavy browsing to rate versus light browsing to rate	−0.002 (−0.20, 0.20)	−0.29 (−0.59, 1.17)	**−0.40 (−0.75, −0.05)**

Table 6 continued

Parameter	Estimates (low 95% CI, up 95% CI)		
	AD	RD	HT
Heavy browsing to rate versus moderate browsing to rate	−0.05 (−0.20, 0.09)	0.01 (−0.89, 0.91)	−0.18 (−0.49, 0.12)
Interaction between basal area and glyphosate rate	−0.00 (−0.00, 0.00)	−0.02 (−0.06, 0.02)	−0.02 (−0.04, 0.004)

Note that elements in bold indicate an effect of that explanatory variable on response variable, and the degree of closeness of 95% CI to mean measures the magnitude of the effect

The interaction between overstory basal area and browsing damage classes has no effect on response variables

2004). The absolute and relative abundance of other hardwoods (dominated by red maple and yellow birch) have also decreased with increasing glyphosate rate, as well as with greater surfactant concentrations (Fig. 2). Increasing surfactant concentration, which was found to have a weak effect through year three of this study (Nelson and Wagner 2011), was correlated with reduced beech density at year nine of our study. However, the strength of the effect was much smaller than the effect of glyphosate rate.

Role of browsing damage

We found that browsing damage at all study sites nine years after herbicide treatment influenced all regenerating tree species. However, sugar maple regeneration was more heavily browsed than other species, which adversely affected the recruitment of sugar maple relative to other species (Fig. 3). Height growth is important during seedling and sapling stages to recruit smaller trees into the larger size classes (Messier et al. 1999; Poorter et al. 2005), and maintain a dominant position relative to neighbors, facilitating eventual recruitment into the overstory before gap closure occurs (Hibbs 1982). Although glyphosate treatment increased the density of sugar maple, selective browsing of sugar maple over beech by ungulates (if persistent over the coming years) will likely reduce the probability of sugar maple achieving understory dominance; thus negating the observed benefits of herbicide release (Horsley et al. 2003; Tripler et al. 2005).

Role of overstory

Besides yellow birch, all major tree species (beech, sugar maple, red maple, balsam fir and red spruce) at our study sites were shade tolerant. These species can survive as advance regeneration for many years under a closed canopy conditions (Canham 1990; Messier et al. 1999). Our results indicated that overstory basal area was the most important explanatory variable in accounting for the variability associated of regeneration height (Table 7), and the effect of overstory basal area was not likely confounded with other explanatory variables (Table 6; Figures S1 and S2 of the Supplementary Materials). Our results indicated that higher levels of overstory basal area reduced the height of sugar maple and other hardwoods, but increased the height of American beech and softwoods; and the effect of overstory basal area was not dependent upon browsing damage. This result is in agreement with our initial hypothesis that the amount of overstory basal area

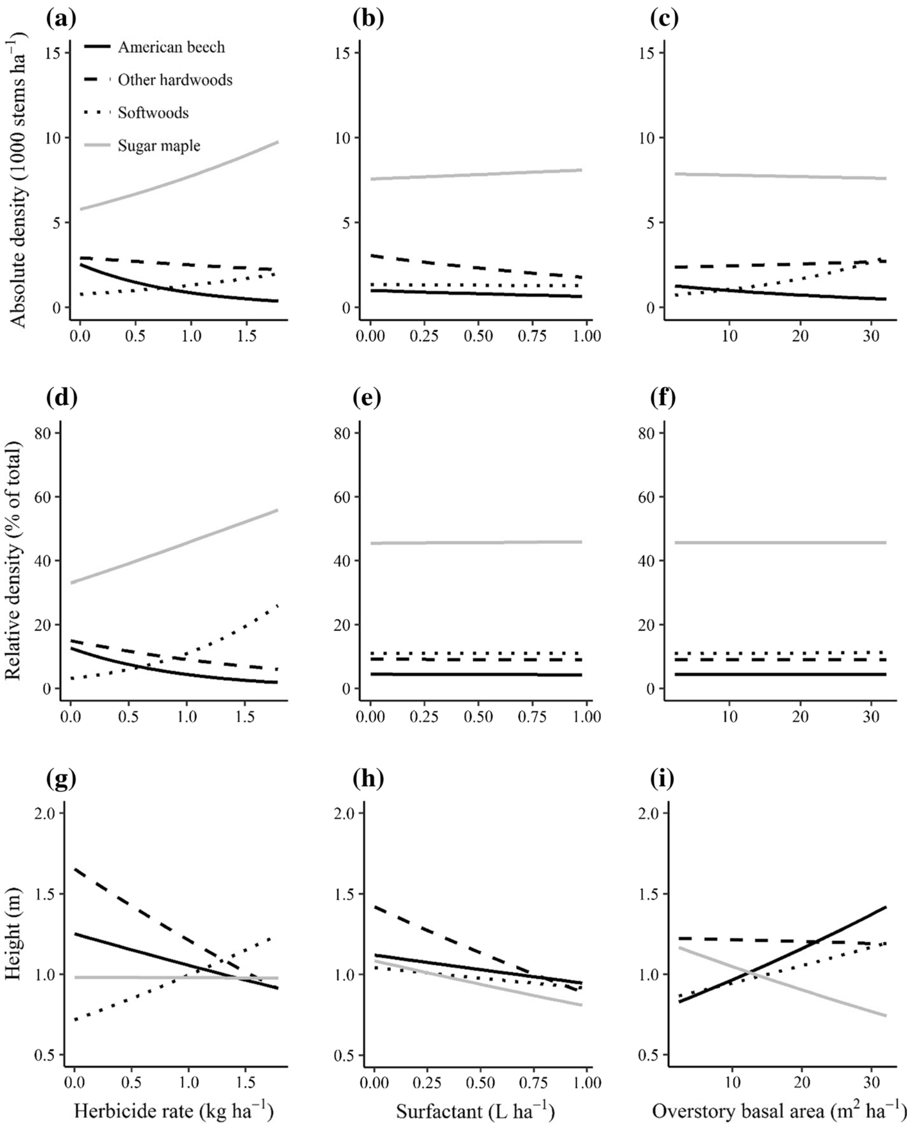

Fig. 2 Predicted average regeneration absolute density (*AD*, stems ha^{-1}), relative density (*RD*, % of total) and height (*HT*, m) with 95% confidence interval of four species groups (American beech, sugar maple, other hardwoods, and softwoods) as a function of herbicide rate, surfactant concentration, and overstory basal area. *Note* Since the pattern of regeneration responses to herbicide rate, surfactant concentration, and overstory basal area were independent of browsing damage classes (heavy, moderate and light), results are presented only for heavily browsed regeneration to avoid redundancy. Differences among three browsing classes for *AD*, *RD* and *HT* are presented in Fig. 3

after treatment would benefit regeneration of beech at the expense of sugar maple, as beech is relatively more shade tolerant than sugar maple (Beaudet et al. 1999, 2007). Although the height of both sugar maple and other hardwoods was reduced with higher overstory

Table 7 Relative contribution to R^2 of explanatory variables in explaining the variability associated with regeneration absolute density (AD), regeneration relative density (RD) and regeneration height (HT)

Predictor variables	Relative contribution to R^2		
	AD	RD	HT
Herbicide rate	0.001	1.267	6.503
Surfactant concentration	0.033	0.287	5.123
Overstory basal area	0.004	0.214	46.431
Browsing classes	22.461	0.734	18.165
Species	42.662	35.070	5.431
Herbicide rate * browsing classes	0.018	1.411	0.799
Overstory basal area * browsing classes	3.738	0.416	6.213
Species * browsing classes	17.237	22.641	1.308
Species * herbicide rate	9.141	26.677	5.631
Species * surfactant concentration	2.527	6.230	0.999
Species * overstory basal area	2.062	3.927	3.052
Herbicide rate * overstory basal area	0.117	1.127	0.345

The interaction term between explanatory variables was indicated by a * sign

basal area, the pattern of reduction was more pronounced for sugar maple than other hardwoods (Fig. 2i). This outcome suggests that sugar maple height growth was reduced more by increased overstory closure than other hardwoods; which is supported by the existing literature (Kobe et al. 1995; Beaudet et al. 1999, 2007). Therefore, a long absence of overstory removal may not favor sugar maple in the recruitment process (Nyland et al. 2006a). The shelterwood seed cuttings at our study sites occurred 11–13 years prior to our measurements, providing a significant amount of time for canopy closure to occur. If this trend continues without subsequent harvest entries, it may eventually favor the dominance of more shade tolerant species including softwoods and beech in the coming years of stand development. However, the observational nature of this analysis and lack of experimental control of browsing suggests a greater need for replication and further evaluation of these relationships.

Conclusion and management recommendations

Our results showed that glyphosate herbicide increased the abundance of sugar maple regeneration in the understory of these mixed hardwood stands, while reducing the abundance of beech. Thus, the initial effect of herbicide treatment observed by Nelson and Wagner (2011) was sustained over a nine-year post-treatment period. However, reduction of browsing pressure and further overstory removal will likely be needed to promote the post-release height growth of sugar maple and reduce the longer-term dominance of beech in these stands. Continued future monitoring at these locations and measurements at additional sites across a wider range of conditions will be necessary to fully understand these dynamics.

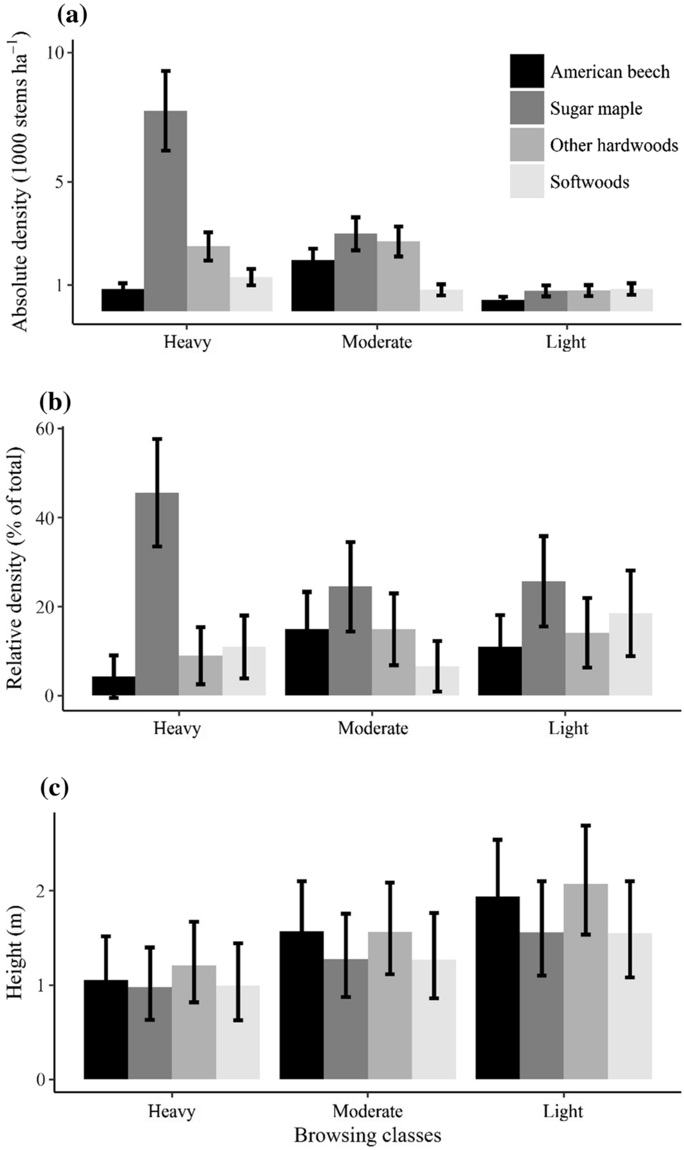

Fig. 3 Predicted regeneration absolute density (*AD*, stems ha^{-1}), relative density (*RD*, % of total), and average height (*HT*, m) of four tree species groups (American beech, sugar maple, other hardwoods and softwoods) for three browsing damage classes (mean ± 95% confidence). **a** Regeneration absolute density, **b** regeneration relative density, **c** regeneration height

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